

Haozhe Lei

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EDUCATION

New York University Ph.D. in Electrical and Computer Engineering <i>Advisor: Prof. Sundeep Rangan, Director, NYU WIRELESS</i> <i>Award: 2023 Ernst Weber Fellowship</i>	Sept. 2022 – Expected May 2027
New York University M.S. in Computer Engineering	Sept. 2020 – May 2022
China Agricultural University B.E. in Electrical Engineering and Automation	Sept. 2015 – Jun. 2019

RESEARCH INTERESTS

- **Methods:** spatially aware RF sensing and wireless localization, ISAC signal processing and inference, probabilistic uncertainty quantification, wireless digital twins and ray tracing, object-centric semantic 3D memory, foundation-model agents, physics-informed and neuro-symbolic RL
- **Systems and Applications:** uncertainty-aware 6G positioning and environment mapping, indoor robot navigation and search, wearable/AR spatial assistance, FR3/mmWave channel-sounding testbeds, multi-band UE antenna/beam management, closed-loop wireless control

RESEARCH EXPERIENCE

NYU WIRELESS <i>Research Assistant</i>	<i>Brooklyn, NY, USA</i> <i>Sept. 2022 – Present</i>
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Research agenda: building spatially aware, uncertainty-aware wireless intelligence that turns sparse RF/visual observations into calibrated spatial beliefs for 6G and robotic decision-making.

- **RF sensing and posterior localization:** Modeling wireless localization as a calibrated spatial posterior over candidate transmitter locations, retaining multimodal hypotheses caused by AoA/SNR ambiguity, receiver directivity, blockage, and reflected paths. Developed *MC-CLE* (Monte Carlo Candidate-Likelihood Estimation), a model-free candidate-posterior method using single-dominant-path AoA/SNR likelihoods with additive evidence accumulation; extended it into *LOCUS-DT* (Localization via Observation-Conditioned Uncertainty Scoring with Digital Twins), which uses a ray-tracing digital twin as a candidate-indexed multipath library and scores measured/synthetic multipath compatibility to address MC-CLE's single-path and site-specific limits, with layout-mismatch diagnostics, cone metrics, and top- k path ablations. Building an FR3/mmWave RFSoc/Pi-Radio channel-sounding testbed with fixed transmitter/receiver baselines, linear-track/D48 pan-tilt sweeps, transmitter mobility on a TurtleBot4 mobile robot, and receiver mobility on a Jackal mobile unmanned ground vehicle (UGV); related manuscripts under review at *IEEE TWC*, *IEEE TSP*, *IEEE GLOBECOM*, and *Asilomar*.
- **Digital-twin priors for wireless-aware navigation:** Developed ray-tracing wireless digital twins and *PIRL* (physics-informed reinforcement learning) methods that use simulated propagation as a transferable prior for zero-shot indoor navigation, wireless SLAM, and symbolic-prior robot policies, with publications in *ICRA*, *IEEE OJ-COMS*, and *RLC (Spotlight)*; extending this line to physical localization/navigation experiments with TurtleBot4 and Jackal UGV platforms, FR3 channel-sounding hardware, and controlled navigation scenarios.
- **UE-centric twins for closed-loop multi-band adaptation:** Developing *MCMB-HDT* (Multi-Cell Multi-Band Handset Digital Twin), a UE-centric framework that couples real urban geometry, base-station

topology, FR1/FR3/mmWave ray tracing, embodied antenna radiation, pedestrian motion, handset pose, and measurement-limited feedback for learning UE-side closed-loop array/band activation. Developed **Transformer**-based rate prediction from sparse asynchronous channel-quality/rate histories and PPO-based power-aware array/band activation under rate, power, and exploration constraints; accepted at *IEEE ICC* and under review at *IEEE JSAC*.

- **Object-centric memory for multimodal spatial reasoning:** Developing lightweight semantic 3D memory for wearable and embodied spatial intelligence, converting sparse egocentric RGB/depth/pose observations into localized object records by fusing VLM-guided object selection, YOLO-World detections, NanoSAM masks, DepthPro geometry, CLIP/DINOv3 embeddings, and cross-view association. The system targets queryable object memory for view retrieval, object localization, instance clustering, spatial search, and agentic robot grounding, with edge/cloud partitioning and resource-aware map updates; ongoing project.

NYU C2SMARTER Center

Research Assistant

Brooklyn, NY, USA

Sept. 2023 – Mar. 2026

- **Traffic mobility control:** Predictive correlated online learning and distributed pan-tilt-camera control for real-time traffic monitoring, estimation, and adaptive coverage; published in *TR-C*.
- **Digital-twin closed loop:** Integration of loop detectors, video analytics, and control feedback for driver-risk-aware mobility/safety prediction and incident response; validated on a Brooklyn urban testbed, with work accepted in *IEEE T-ITS*.
- **Distributed online control:** Distributed online convex optimization for networked sensing/control systems with nonseparable costs and time-varying constraints; accepted in *IEEE L-CSS*.

NYU Center for Cybersecurity

Research Assistant

Brooklyn, NY, USA

Sept. 2021 – Jan. 2024

- **Safe adaptive control:** Game-theoretic, neuro-symbolic, and meta-RL methods for nonstationary driving, level- k pedestrian-aware autonomy, sampling-attack robustness, and distributed adaptive penetration testing; publications in *ICRA* and *MILCOM*.
- **Route-recommendation control under strategic behavior:** Incentive-compatible route recommendation and proactive risk assessment for misinformation/demand attacks in navigation systems, combining Wardrop-equilibrium reasoning, Stackelberg-game modeling, and robust control; related work published in *IEEE TIFS*.

PUBLICATIONS

Legend: **bold** = Haozhe Lei, * co-first author, † sole corresponding author.

Journal Articles

- [J1] **Haozhe Lei**[†], Roberto Bomfin, Marwa Chafii, and Sundeep Rangan. “Site-Agnostic Posterior Inference for Indoor Localization with Ray-Tracing Wireless Digital Twins”. *IEEE Transactions on Wireless Communications* (2026). Under review.
- [J2] **Haozhe Lei**[†], Hao Guo, and Sundeep Rangan. “Learning a Measurement-to-Posterior Map for Wireless Localization”. *IEEE Transactions on Signal Processing* (2026). Under review.
- [J3] Ruibin Chen*, **Haozhe Lei**^{*,†}, Marco Mezzavilla, Hitesh Poddar, Tomoki Yoshimura, and Sundeep Rangan. “MCMB-HDT: A Multi-Cell Multi-Band Handset Digital Twin for Learning-Based Closed-Loop Array Activation”. *IEEE Journal on Selected Areas in Communications* (2026). Under review.
- [J4] Zhaoye Pan, **Haozhe Lei**, Fan Zuo, Zilin Bian, and Tao Li. “Distributed Online Convex Optimization with Nonseparable Costs and Constraints”. *IEEE Control Systems Letters* 10 (2026), pp. 391–396. [\[PDF\]](#).
- [J5] Tao Li*, Zilin Bian*, **Haozhe Lei***, Fan Zuo*, Ya-Ting Yang*, Quanyan Zhu, Zhenning Li, Zhibin Chen, and Kaan Ozbay. “Digital Twin-Based Driver Risk-Aware Predictive Mobility Analytics for Real-Time Situational Awareness Through Cooperative Sensing”. *IEEE Transactions on Intelligent Transportation Systems* (2025). [\[Article\]](#).

- [J6] Tao Li, **Haozhe Lei**, Hao Guo, Mingsheng Yin, Yaqi Hu, Quanyan Zhu, and Sundeep Rangan. “Digital Twin-Enhanced Wireless Indoor Navigation: Achieving Efficient Environment Sensing with Zero-Shot Reinforcement Learning”. *IEEE Open Journal of the Communications Society* (2025). [\[Article\]](#).
- [J7] Ya-Ting Yang, **Haozhe Lei**, and Quanyan Zhu. “PRADA: Proactive Risk Assessment and Mitigation of Misinformed Demand Attacks on Navigational Route Recommendations”. *IEEE Transactions on Information Forensics and Security* (2025). [\[PDF\]](#).
- [J8] Zilin Bian, Tao Li, **Haozhe Lei**, Fan Zuo, Ya-Ting Yang, Quanyan Zhu, Zhenning Li, and Kaan Ozbay. “Multi-level Traffic-Responsive Tilt Camera Surveillance through Predictive Correlated Online Learning”. *Transportation Research Part C: Emerging Technologies* (2024). [\[Article\]](#).

Conference Papers

- [C1] **Haozhe Lei**[†], Roberto Bomfin, Marwa Chafii, and Sundeep Rangan. “LOCUS-DT: Localization via Observation-Conditioned Uncertainty Scoring with Digital Twins”. *IEEE GLOBECOM*. Under review. 2026.
- [C2] **Haozhe Lei**[†], Hao Guo, Tommy Svensson, and Sundeep Rangan. “Beyond Point Estimates: Likelihood-Based Full-Posterior Wireless Localization”. *Asilomar*. Under review. 2026. [\[PDF\]](#).
- [C3] Mingsheng Yin^{*}, Tao Li^{*}, **Haozhe Lei**^{*}, Yaqi Hu, Sundeep Rangan, and Quanyan Zhu. “Zero-Shot Wireless Indoor Navigation through Physics-Informed Reinforcement Learning”. *IEEE ICRA*. 2024. [\[PDF\]](#).
- [C4] Tao Li^{*}, **Haozhe Lei**^{*}, and Quanyan Zhu. “Self-Adaptive Driving in Nonstationary Environments through Conjectural Online Lookahead Adaptation”. *IEEE ICRA*. 2023. [\[PDF\]](#).
- [C5] Ruibin Chen^{*}, **Haozhe Lei**^{*,†}, Hao Guo, Marco Mezzavilla, Hitesh Poddar, Tomoki Yoshimura, and Sundeep Rangan. “Transformer-Based Rate Prediction for Multi-Band Cellular Handsets”. *IEEE ICC Workshops*. 2026. [\[PDF\]](#).
- [C6] Roberto Bomfin, **Haozhe Lei**, Sundeep Rangan, and Marwa Chafii. “Likelihood-Based Wireless Localization with Last-Bounce Spatial Features”. *Asilomar*. Under review. 2026.
- [C7] Tao Li, **Haozhe Lei**, Mingsheng Yin, and Yaqi Hu. “Reinforcement Learning with Physics-Informed Symbolic Program Priors for Zero-Shot Wireless Indoor Navigation”. *RLC*. Selected as a spotlight paper. 2025. [\[PDF\]](#).
- [C8] **Haozhe Lei**, Yunfei Ge, and Quanyan Zhu. “ADAPT: A Game-Theoretic and Neuro-Symbolic Framework for Automated Distributed Adaptive Penetration Testing”. *IEEE MILCOM*. 2024. [\[PDF\]](#).

TEACHING & MENTORING

New York University

Spring 2024 – Present

Graduate Teaching

- ECE-GY5213 Introduction to System Engineering, Teaching Assistant Spring 2024

Graduate Mentoring

- **Ruibin Chen**, Ph.D. student, New York University Feb. 2025 – Present
Project: Multi-Band Handset Digital Twins and Closed-Loop Array/Band Activation (IEEE ICC accepted; IEEE JSAC under review)
- **Yuheng Liu**, M.S. student, New York University Jan. 2026 – Present
Project: Multimodal Spatial Reasoning with Scene-Graph Memory and Agentic Inference

INTERNSHIPS

Astrabeam

New York, NY, USA

Research Intern

Jun. 2025 – Aug. 2025

- Doppler-based mmWave radar processing in ROS2 for odometry & object detection.

- CenterNet (PIoU) rotational detection; on-site deployment to manufacturing lab.

AWARDS

- Ernst Weber Fellowship (NYU Tandon School of Engineering), 2023
- Tandon Research Excellence Exhibit [\[Link\]](#), Project: *AI-Driven Interactive Safe Autonomous Driving*, 2022
- School of Engineering Fellowship (NYU Tandon School of Engineering), 2022
- China National University Student Innovation & Entrepreneurship Training Program, 2017-2019

TECHNICAL EXPERTISE

- **Programming & Tools:** Python, Linux/Shell scripting, MATLAB, R, LaTeX, Kubernetes (HSRN cluster)
- **Simulation & Frameworks:** ROS2, Sionna, Wireless InSite, SUMO & CARLA (Transportation)
- **Robotics & Wireless Hardware:** TurtleBot4 mobile robot, Jackal mobile unmanned ground vehicle (UGV), RFSoc/Pi-Radio, D48 pan-tilt unit, linear track, TI mmWave radar (AWR1642, AWR6843), DCA1000

REVIEW SERVICES

Journals

- IEEE Transactions on Signal Processing (TSP)
- IEEE Transactions on Communications (TCOM)
- IEEE Wireless Communications Letters (WCL)
- IEEE Vehicular Technology Magazine (VTM)
- IEEE Control Systems Letters (L-CSS)

Conferences

- IEEE International Conference on Communications (ICC 2026)
- IEEE International Conference on Computer Communications (INFOCOM 2025)
- IEEE Global Communications Conference (GLOBECOM 2025)
- IEEE Conference on Decision and Control (CDC 2025)
- IEEE International Conference on Robotics and Automation (ICRA 2024)
- American Control Conference (ACC 2024)

REFERENCES

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